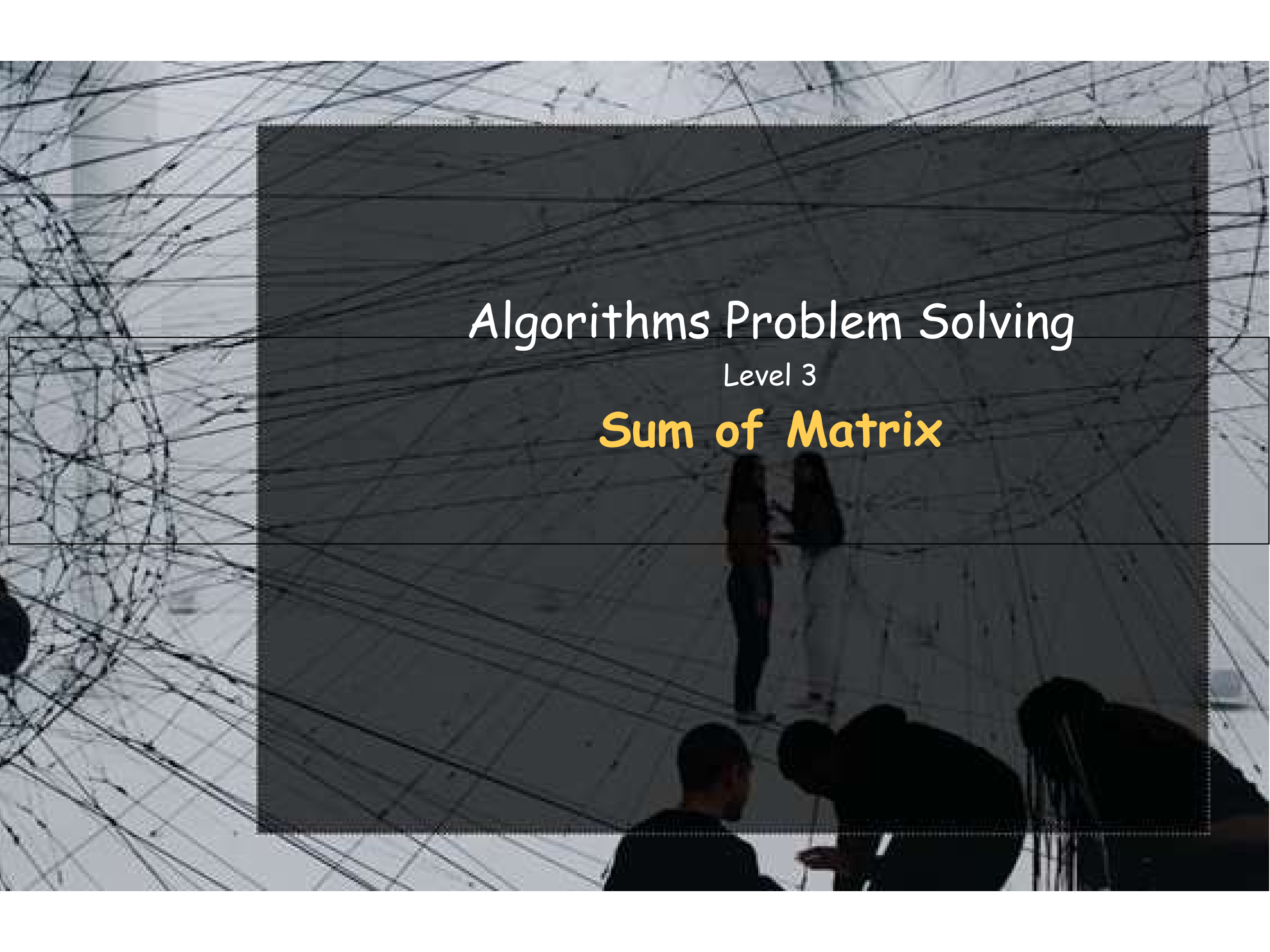


Algorithms Problem Solving

Level 3

Sum of Matrix



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Definition

Vowel Frequency Counter

-  Program counts how many vowels appear in a complete text string.
-  The standard English vowel set used here is `a`, `e`, `i`, `o`, and `u`.

Explanation

Vowel Predicate

-  `IsVowel()` is used as a reusable helper function.
-  It determines whether a character belongs to the vowel set.

String Traversal

-  The program examines every character in the input string.
-  Each character is passed to the vowel-checking function.

Frequency Accumulation

-  The counter starts at `0`.
-  It increases whenever a vowel is detected.

Procedures

1. Define `IsVowel(char)`.
2. Read the complete input string.
3. Traverse every character.
4. Call `IsVowel()` for each character.
5. Increment the counter for every vowel.
6. Display the total.

💡 Example

Input:

Sara Alterkawi

➡ Result

Number of vowels is: 6

Note

- ◆ The helper can normalize character case before checking.
- ◆ The result counts occurrences, not distinct vowel types.
- ◆ A vowel appearing several times is counted several times.

⊘ Warning

- ◆ ``cin >> str`` reads only the first word and therefore should not be used when processing complete sentences.
- ◆ Consonants, spaces, digits, and punctuation must not increase the vowel counter.
- ◆ Do not forget to handle uppercase vowels if the input may contain capital letters.



Comparison



`CountVowels()`

- ✓ Counts the total number of vowels.
- ✓ Returns a numerical frequency.
- ✓ Traverses the complete input.



`PrintVowels()`

- ✓ Displays the actual vowel characters.
- ✓ Preserves their occurrence order.
- ✓ Does not primarily calculate a frequency.

✦✦ Summary

- ✓ Scans a complete string for vowel characters.
- ✓ Reuses `IsVowel()` as a boolean predicate.
- ✓ Counts every vowel occurrence.
- ✓ Demonstrates reusable filtering and counting logic.

Thank you 😊